FOEZ AHMED

Lead RTL Design & Verification Engineer

RISC-V SoC | AXI Interconnect | Cache Coherence | DDR Subsystems

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PROFESSIONAL SUMMARY

RTL Design & Verification Engineer with 4.5+ years of experience spanning RISC-V SoC architecture, AXI interconnects, cache coherence, and DDR controller development. Technical lead for a quad-core RV64GV SoC targeting 1.5 GHz on 12 nm, with architecture and verification ownership through tape-out. Built a reusable class-based AXI4 Master/Slave VIP and owned a custom directory-based coherent cache protocol end to end. Currently leading DDR controller microarchitecture and AXI integration for a ~50-engineer team.

TECHNICAL SKILLS

Architecture & Design: RV64G cores, RISC-V SoC architecture, AXI / AHB / APB / OBI / Wishbone, AXI ID/data-width conversion, DDR controller microarchitecture, directory-based cache coherence, cache design with foundry memory blocks, clock/reset generation with foundry PLL and standard cells

Peripherals: UART, I2C, I2S, SPI

Verification: SystemVerilog, SVA, class-based testing, directed testing, functional coverage, UVM, CoCoTB, assembly/C system-level testing, mixed-language simulation

Emulation & Timing: Multi-FPGA emulation, Xilinx Vivado, static timing analysis

Tools: Synopsys VCS, Siemens Questa, Xilinx Vivado, Icarus Verilog, Git

Programming & Scripting: C, Python, Bash, GNU Make

PROFESSIONAL EXPERIENCE

Lead Engineer, RTL Design & Verification

Jan 2026 – Present

ADN Semiconductors, Mohakhali, Dhaka

- Define the microarchitecture and data-handling architecture of a custom DDR controller IP and its AXI integration into the SoC.
- Provide technical direction for a team of approximately 50 engineers; lead architectural, functional, and RTL/design reviews.
- Guide optimization of RTL implemented by junior engineers from the microarchitecture specification.
- Contribute to RISC-V-related projects and FPGA emulation activities.

Engineer, RTL Design & Verification

Jan 2022 – Jan 2026

Dynamic Solution Innovators Ltd., Mohakhali DOHS, Dhaka

Client: California-based fabless semiconductor company (client/project names withheld under NDA)

RISC-V Quad-Core SoC (taped out; 12 nm, 1.5 GHz target)

- Technical lead for architecture and verification of a 4× RV64GV SoC with a dedicated wireless communication/broadcasting subsystem, 16 KB 4-way L1 I-cache and 32 KB 4-way L1 D-cache per core, and a 2 MB 4-way shared L2 cache, working with 30+ engineers (10–12 design, 16–20 verification, ~5 firmware).
- Reviewed client design and verification deliverables, designed peripheral RTL, and personally owned interconnect architecture definition, debugging, and fixes.
- Co-proposed with the client, then implemented and optimized, a split of the AXI fabric into a low-latency, high-bandwidth compute network with caching and a system network for peripherals and less performance-sensitive traffic.
- Resolved AXI ID-width growth across the interconnect hierarchy by defining ID-width conversion points, added data-width conversion where required, and revalidated the fabric with the existing AXI verification suite after the interconnect implementation changed from the original AXI crossbar during silicon implementation.
- Established the multi-stage system verification flow, using assembly and C tests:
  - Smoke: IP boot, clock and reset propagation.
  - Per-core ISA functional: supported RISC-V features/extensions, ~100% functional coverage.
  - Interconnect/system: AXI configuration and data propagation, ~95% functional coverage.
  - Stress: varied system and interconnect load scenarios, ~90% functional coverage.

  - Benchmark: floating-point matrix multiplication and multi-master reads/writes to shared and distinct memory locations, comparing execution time and throughput across configurations.
- Built the initial two-FPGA emulation framework for a design that exceeded a single FPGA, including a virtual inter-FPGA communication interface over shared host memory for bidirectional synchronous transactions.

AMBA AXI4 Master/Slave Verification IP

- Developed and verified a SystemVerilog class-based (non-UVM) AXI4 Master/Slave VIP supporting simultaneous bidirectional transactions, independent read/write channels, back-to-back transfers, all burst modes, exclusive access, and QoS reordering, with comprehensive protocol-compliance checking.
- Designed driver/monitor-style interfaces for straightforward integration into UVM environments, a simplified programming interface, and an extensible functional coverage model covering AXI behavior and memory-related options.
- Reused across multiple AXI-related projects.

Custom Directory-Based Coherent Cache System

- Owned the custom (non-AXI) coherence communication protocol end to end, from definition through verification.
- Architected 4-way private caches with directory-based coherence tracking logical ownership of memory regions (no physical partitioning) and write-back to main memory on eviction.
- Translated customer requirements into the microarchitecture and implementation plan; split RTL/microarchitecture and verification work roughly equally, including a dedicated environment to verify coherence across the private caches.

EDUCATION

B.Sc. in Electrical and Electronic Engineering Feb 2022
Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh

SELECTED PUBLICATIONS

- Comparative Performance Analysis of TMD based Multi-Bridge Channel Field Effect Transistor. IEEE NAP, Nov 2020.
- A Comprehensive Study and Modern Take on Intel 8086 like Modular Trainer Board System. IEEE RAAICON, Dec 2021.

8 peer-reviewed publications in total across IEEE and other international conferences; full list available on request.

OPEN SOURCE

github.com/foez-ahmed